



Linux Package Management Notes

Topic: **apt update** vs **apt upgrade**

1 **sudo apt update**

👉 What does it do?

- Refreshes the **package list**.
- Fetches the latest package information from the internet
- Does not install or update any software
- It only checks:
- Checks whether new versions are available or not.

✅ In Simple Words

The system checks:

“Is a new version available?”

⚠️ It **does NOT install or update** any software.

🧠 Example Command

```
sudo apt update
```

📱 Real-Life Example

Like refreshing the **Play Store**:

- You check updates
 - But apps are NOT updated yet
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✅ Result After Running

System becomes aware that:

- New version of **nginx** is available
- New version of **git** is available
- Other packages may have updates

❌ Nothing is installed automatically.

2 **sudo apt upgrade**

👉 What does it do?

- Upgrades already installed packages to newer versions.

✅ In Simple Words

- Updates installed software
- Installs latest versions available

✓ What it DOES:

- Updates existing packages

✗ What it DOES NOT do:

- Does not install new packages
- Does not remove old packages

🧠 Example Command

sudo apt upgrade

📱 Real-Life Example

Like clicking:

Play Store → Update All Apps

3 Why Use Both Together?

Command:

sudo apt update && sudo apt upgrade

🔍 Reason (Step-by-Step)

Step 1 — apt update

System learns:

“New versions are available.”

Step 2 — apt upgrade

System installs:

“Now update all installed packages.”

✓ Workflow Diagram

apt update



Check new versions

apt upgrade



Install updates



2>&1 — Quick Revision

✓ Linux Output Types

- 1 (STDOUT) → Normal output
 - 2 (STDERR) → Error output
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✓ What is 2>&1 ?

👉 Send **errors** to the same place as **normal output**.

Simple meaning:

Output + Error = Together

✓ Example

command > log.txt 2>&1

- ✓ Normal output → log.txt
- ✓ Error output → log.txt

Both saved in one file.

✓ Without 2>&1

Errors show on screen ❌

✅ **With 2>&1**

Everything saved together ✅

✅ **Memory Trick**

1 = output

2 = error

2>&1 → error goes to output

★ **One line:**

2>&1 = combine output and error in same place.